

Methods for Overround Correction and Probability Inference in Betting Markets

MATH3009: Individual Project Report (20cp)
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Contents

1	Introduction	4
1.1	The Favourite-Longshot Bias	5
2	Rescaling Methods	5
2.1	Additive Method	5
2.2	Multiplicative Method	6
2.3	Power Method	6
3	The Shin Method	8
3.1	Introduction to Shin's Method in a Two-Horse Race	8
3.2	Shin's Method for $n \geq 3$ Outcomes	13
3.3	Using Shin's Method for Probability Inference	16
3.4	Proof that Shin's and Additive Methods are Equivalent for $n=2$ Outcomes	17
4	Regression Methods	18
4.1	Linear Regression	18
4.2	Polynomial Regression	20
4.3	Logistic Regression	21
4.4	Ordered Logistic Regression	22
5	Data	23

5.1	Data for $n = 2$ Outcomes	23
5.2	Data for $n \geq 3$ Outcomes	24
6	Proper Scoring Rules for Assessing Probabilistic Accuracy	24
6.1	Brier Score	25
6.2	Log-Loss Score	26
6.3	Ranked Probability Score	27
7	Results	28
7.1	Results on $n = 2$ Outcomes	28
7.2	Results on $n \geq 3$ Outcomes	29
8	Conclusion	31

Acknowledgement of Sources

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Signed James Restell
Date 28/02/2026

1 Introduction

Betting markets are widely considered to be the most accurate source of publicly-available probability forecasts in sports [15]. The advanced quantitative modelling techniques used by bookmakers along with the ideas of the Wisdom of Crowds [14] and the Efficient Market Hypothesis [6] mean that the odds provided by bookmakers represent an incredibly well-informed probability forecast of any given sporting event. For any individual concerned with modelling sporting outcomes, whether through a sports betting lens or from a performance analysis point of view, this source of information is highly valuable.

However, the odds offered by any bookmaker do not correspond to valid probabilities. This presents an obstacle between sports modellers and this valuable source of information. In this report we will discuss and compare various methods of tackling this challenge in the context of match outcome betting. In particular we will focus on sports with 2 outcomes (e.g. Darts, Tennis, Cricket) and 3 outcomes (e.g. sports with draws allowed such as Football).

Firstly let us define the following notation for values that we frequently use across sections:

- o_i - the decimal odds offered by a bookmaker on outcome i occurring. This corresponds to the factor that your stake is multiplied by should outcome i occur, i.e. if you stake £4 on a bet at odds of 2.50, you will receive £4 × 2.50 = £10 should your bet win (your original £4 stake plus £6 profit). For readers who are more familiar with fractional odds (e.g. 4/1, 5/2, 1/4 etc.), evaluating these fractions and adding 1 will leave you with the equivalent decimal odds. $\mathbf{o} := (o_1, \dots, o_n)$ represents the complete market odds on an event with n possible outcomes.
- p_i - the “true” probability of outcome i occurring according to the bookmaker. This is the value we are trying to infer based on the market prices $\mathbf{o}$. We may add a label to these probabilities to indicate which method we have used to obtain them. We refer to the full probability distribution of outcomes as $\mathbf{p} = (p_1, \dots, p_n)$.
- $\beta := \sum_{i=1}^n 1/o_i$ - the **booksum** of the market $\mathbf{o}$. This represents the sum of the implied probabilities of the market. If the odds represent valid probabilities then $\beta = 1$, however we know that in practice $\beta > 1$ for all sports and all bookmakers. We refer to the value $(\beta - 1)$ as the **overround**.
- The “1X2” market is the market for betting on the outcome of a football match, where 1 corresponds to the home team winning, X corresponds to a draw, and 2 corresponds to the away team winning.

Using the idea of expected value we can see that if a bookmaker is offering odds o_i on an outcome, the bookmaker must believe that the corresponding probability p_i satisfies

$$\frac{1}{o_i} \geq p_i$$

otherwise the bookmaker will make a loss in the long term. Equally, a bettor placing a bet at odds of o_i must believe that the true probability of the outcome p_i satisfies

$$\frac{1}{o_i} \leq p_i$$

We can see that a “fair bet” occurs when $1/o_i = p_i$. Famously, bookmakers are not primarily concerned with offering fair bets, and in order to ensure their business remains profitable will often add a built-in margin to their odds. For most bookmakers, this booksum will lie between 1.05 and 1.2 on match outcome markets, and this represents a 5 – 20% overround that the bookmaker has over a bettor (this is also commonly referred to as “vig”, “edge” or “margin”). The rest of this report will discuss the various methods of removing this overround to solve the following central problem:

As a bettor being offered a market of odds $\mathbf{o} = (o_1, \dots, o_n)$ with booksum $\beta > 1$, how can I best infer the valid probability distribution $\mathbf{p} = (p_1, \dots, p_n)$ satisfying $\sum_i p_i = 1$ that the bookmaker likely used to set their odds?

1.1 The Favourite-Longshot Bias

One of the central ideas in this report is the **favourite-longshot bias**, first observed in horse-racing markets by Griffith (1949) [8]. This phenomenon sees betting odds for favourites priced much closer to their fair price than odds for longshots. There are differing views on the cause of this discrepancy, but from a psychological point of view we may suspect that bettors are risk-loving and consistently overestimate the likelihood of outsiders winning, and underestimate the likelihood of favourites winning. This in turn sees bettors willing to place bets on longshots at much lower odds and so a bookmaker will tend to apply margin disproportionately to the odds of the outsider winning. Cain et al. (2003) [2] found that this phenomenon is present in a variety of sports markets, but that the pattern of bias in the odds differs between sports and markets.

Mathematically speaking, this means that when we are attempting to map implied probabilities $q_i = 1/o_i$ to true latent probabilities p_i , we are looking for a function $f : [0, 1] \rightarrow [0, 1]$ that is convex in terms of q_i .

In Section 3 we see another explanation for why the favourite-longshot bias exists in betting odds, which is that it is the optimal response from a bookmaker when trying to negate the impact of insider's in the market on their profit [12].

2 Rescaling Methods

In this section we present three related methods for removing overround, the Additive, Multiplicative, and Power methods. While the Additive and Multiplicative methods are simpler, they do not account for favourite-longshot bias whereas the slightly more complex Power method does.

2.1 Additive Method

The additive method works by assuming that overround has been split evenly between the n possible outcomes[3]. Thus we find probabilities via

$$p_i^{(A)} = \frac{1}{o_i} - \frac{\beta - 1}{n} \quad \text{for } i = 1, \dots, n$$

We can prove that these probabilities sum to 1 as follows

$$\begin{aligned} \sum_{i=1}^n p_i^{(A)} &= \sum_{i=1}^n \left\{ \frac{1}{o_i} - \frac{\beta - 1}{n} \right\} \\ &= \sum_{i=1}^n \left\{ \frac{1}{o_i} \right\} - \beta + 1 \\ &= \beta - \beta + 1 \\ &= 1 \end{aligned}$$

A well-known flaw of the additive method is that it can produce negative probabilities for outsiders or probabilities greater than 1 for strong favourites depending on the value of the overround. We can see that the former occurs when

$$\begin{aligned} \frac{1}{o_i} - \frac{\beta - 1}{n} &< 0 \\ \frac{1}{o_i} &< \frac{\beta - 1}{n} \\ n &< o_i(\beta - 1) \end{aligned}$$

For example if we consider a Football match with $\mathbf{o} = (1.1, 5, 25)$, we get an overround of roughly 15% and we would predict $p_3^{(A)} = 1/25 - 0.15/3 = -0.01$ which is invalid. These scenarios only occur with unusually long/short odds and high overround so the method is still useful in most normal settings.

2.2 Multiplicative Method

The multiplicative method works in a similar way by assuming overround has been split proportionally between the outcomes. In this case we get that

$$p_i^{(M)} = \frac{1}{o_i \beta}$$

These probabilities will sum to 1 as

$$\begin{aligned} \sum_{i=1}^n p_i^{(M)} &= \sum_{i=1}^n \frac{1}{o_i \beta} \\ &= \frac{1}{\beta} \sum_{i=1}^n \frac{1}{o_i} \\ &= \frac{\beta}{\beta} \\ &= 1 \end{aligned}$$

The main drawback of this method is that it fails to account for the favourite-longshot bias. Let $q_i = 1/o_i$ be the implied probability of the bookmaker's odds, and consider the ratio of the true probabilities

$$\frac{p_i^{(M)}}{p_j^{(M)}} = \frac{1/(\beta o_i)}{1/(\beta o_j)} = \frac{1/o_i}{1/o_j} = \frac{q_i}{q_j}$$

and so we can see that the multiplicative method preserves the ratios of implied probabilities with true probabilities which is not consistent with a method that accounts for favourite-longshot bias.

2.3 Power Method

In the power method, we raise prices to a constant power k in order to obtain $p_i^{(P)}$:

$$p_i^{(P)} = \left(\frac{1}{o_i} \right)^k$$

where we will use an iterative method to find k whilst ensuring that $\sum_i p_i = 1$. We can define

$$f(k) = \sum_{i=1}^n \left(\frac{1}{o_i} \right)^k - 1$$

and use a numerical technique such as Newton-Raphson to find the value of k^* such that $f(k^*) = 0$. Then we can find our adjusted probabilities using

$$p_i^{(P)} = \left(\frac{1}{o_i} \right)^{k^*}$$

In order to use the Newton-Raphson method to find the root of this function, we need to define the derivative f' . We use the fact that $\frac{d}{dx} a^x = a^x \cdot \ln a$ to find

$$f'(k) = \sum_{i=1}^n \{ (o_i)^{-k} \cdot \ln(o_i^{-1}) \}$$

Algorithm 1: Newton-Raphson Algorithm to Find k^*

Data: $\mathbf{o} = (o_1, \dots, o_n)$
Result: k^*
 $x_0 \leftarrow 1$;
 $x_n \leftarrow x_0$;
 $x_{n+1} \leftarrow x_n - \frac{f(x_n)}{f'(x_n)}$;
 $\epsilon \leftarrow 1e-8$;
while $|x_{n+1} - x_n| > \epsilon$ **do**
 $x_n \leftarrow x_{n+1}$;
 $x_{n+1} \leftarrow x_n - \frac{f(x_n)}{f'(x_n)}$;
 $k^* \leftarrow x_{n+1}$

We now use the following algorithm to find k^* which we can use to find $\mathbf{p}^{(P)}$.

For example on an evenly matched football match, a bookmaker may quote odds of (2.7, 3, 2.8) for the outcomes {Home Win, Draw, Away Win}. These odds have a roughly 5.7% overround. We can use the algorithm above to find that, in this scenario, $k^* \approx 1.0569$ and we get the probabilities

$$\mathbf{p}^{(P)} = (0.350, 0.313, 0.337)$$

which sum to 1.

This is a popular method for removing overround as it guarantees that probabilities will lie in the interval $[0, 1]$, because we can assume that $o_i > 1$ (no bets can return less than the stake), and so we know that $(1/o_i) \in (0, 1)$ and raising this to a positive power k^* means the resulting probability $p_i^{(P)} \in (0, 1)$. In fact we know that $k^* > 1$ by assuming that $\beta > 1$ for any real bookmaker. Considering the function $f(k)$ above, we can see that

$$f(1) = \sum_{i=1}^n \frac{1}{o_i} - 1 = \beta - 1 > 0$$

We can also see that since $1/o_i < 1$, $(1/o_i)^k$ will approach zero for large k . So

$$\lim_{k \rightarrow \infty} f(k) = -1$$

This means that, by the intermediate value theorem, the root of $f(k)$, must lie in the interval $k^* \in (1, \infty)$.

Another strength of the power method is that it does adjust for the favourite-longshot bias. We can show this by comparing the ratios of true probabilities and implied probabilities again.

$$\frac{p_i^{(P)}}{p_j^{(P)}} = \frac{q_i^k}{q_j^k} = \left(\frac{q_i}{q_j} \right)^k$$

Suppose that outcome i is the favourite and outcome j is the longshot, so that $q_i/q_j > 1$. Since $k > 1$ we can say that

$$\frac{p_i^{(P)}}{p_j^{(P)}} = \left(\frac{q_i}{q_j} \right)^k > \frac{q_i}{q_j}$$

So we have increased the difference in probability between the favourite and the longshot, which is exactly the favourite-longshot bias correction.

3 The Shin Method

In this section we consider the problem of setting odds from the perspective of the bookmaker. We introduce Shin's method which examines how a bookmaker can set odds in a market where she knows that some of the bettors have more information than them. We outline two hypothetical scenarios a bookmaker may face and show the solutions to the following problem:

If I (a bookmaker), know the true underlying probabilities of outcomes $\mathbf{p} = (p_1, \dots, p_n)$, but I also know that some proportion z of the money being staked comes from bettors who have more information than me, how can I optimally set my prices $\pi = (\pi_1, \dots, \pi_n)$?

We finally show how we can reverse engineer what a bookmaker must think about the true probabilities of events based on the odds they have set.

3.1 Introduction to Shin's Method in a Two-Horse Race

Shin (1991)[11] asks us to consider a bookmaker at a racetrack who is offering bets to a group of gamblers, some of whom have insider knowledge about the outcome of the race. This race is a two-horse race (we will expand to the 3 outcome scenario relevant to the 1X2 market later). The bookmaker cannot distinguish between the bettors with insider knowledge and the ones without, but has some idea of the proportion of gamblers that have insider knowledge. Shin's method examines how the bookmaker should optimally set their odds in this scenario.

Let's examine the three entities in this model:

- (1) **The Bookie** - a monopolistic bookmaker who is setting the odds on the race. The Bookie knows the probability distribution of possible results of the race, but doesn't know which outcome will be realised.
- (2) **The Insiders** - this group has insider information about the outcome of the race. They know which outcome will be realised with certainty. Since all Insiders behave the same we refer to them collectively as the **Insider**.
- (3) **The Outsiders** - a continuum of uninformed gamblers who place bets according to their own subjective opinions of the probability distribution of outcomes.

We define the following values:

- (π_1, π_2) are the tickets being traded by the bookmaker. π_1 is the price of the ticket that pays $\mathcal{L}1$ if horse 1 wins, and likewise for π_2 . These prices are **not** decimal odds, they are the equivalent of a bet at $1/\pi_i$ in decimal odds.
- $\mathcal{L}z$ is the wealth of the Insider.
- We index the Outsiders using $q \in [0, 1]$, where the Outsider with index q believes that Horse 1 will win with probability q and Horse 2 will win with probability $1 - q$.
- The outsiders collectively have wealth of $\mathcal{L}y$, and the distribution of wealth is uniform over $[0, 1]$. We can assume without loss of generality that the total wealth of our gamblers is 1, so that $y = 1 - z$.

We will assume that all of our Outsiders are risk-neutral, and so if a bookmaker posts prices of (π_1, π_2) they will act according to the following betting strategy:

- If $\pi_1 < q$ and $\pi_2 \geq 1 - q$, then bet on Horse 1 to win.
- If $\pi_2 < 1 - q$ and $\pi_1 \geq q$, then bet on Horse 2 to win.
- If $\pi_1 > q$ and $\pi_2 > 1 - q$, then do not bet.
- If $\pi_1 < q$ and $\pi_2 < 1 - q$, then
 - Bet on Horse 1 to win if $q/\pi_1 > (1 - q)/\pi_2$
 - Otherwise bet on Horse 2.

The scenario described by Shin is broken down into 3 stages.

- Stage 1. “Nature” chooses a value p_1 from a *Uniform*(0,1) distribution. This is the probability of Horse 1 winning and so $p_2 = 1 - p_1$.
- Stage 2. The Bookie can observe this choice of p_1 , and posts its prices (π_1, π_2) according to the restraints $0 \leq \pi_i \leq 1$ for $i = 1, 2$.
- Stage 3. “Nature” takes a sample from a *Bernoulli*(p_1) distribution to determine the winner of the race. The Insider has access to this sample, and is free to bet on the race using her wealth of $\mathcal{L}z$. The Outsiders also place their bets now according to the strategy above.
- Stage 4. The race is run, and the bets are settled according to the prices in *Stage 2*.

What values of (π_1, π_2) can we choose in Stage 2 of this scenario that will optimise the expected profit of a bookmaker given a proportion of bettors z know the outcome of the race with certainty, and the remaining $(1 - z)$ proportion of bettors bet according to their beliefs that Horse 1 will win with probability $q \sim \text{Uniform}(0, 1)$.

We begin with the fact that

$$\pi_1 + \pi_2 \geq 1$$

for all choices of p_1 . Allowing $\pi_1 + \pi_2 < 1$ would make the Bookie a sure loser by the Dutch Book Theorem and so any rational Bookie will not let that happen. In this case bettors would be able to simply buy both tickets and receive a guaranteed profit of $1 - (\pi_1 + \pi_2) > 0$.

We proceed by examining the expected payout the bookmaker receives from the Outsiders. Since Outsiders only bet when $q > \pi_i$, and q is uniformly distributed over $(0, 1)$, we can expect to sell $(1 - \pi_i)$ tickets. We also know that the total wealth of Outsiders $(1 - z)$ is uniformly distributed between the outsiders so we can expect the revenue from sales of ticket π_i to Outsiders to be

$$(1 - z)(1 - \pi_i)$$

If ticket i then wins, we now have to pay the Outsiders who purchased this ticket

$$(1 - z) \left(\frac{(1 - \pi_i)}{\pi_i} \right)$$

Given that ticket i will win with probability p_i , the expected profit on sales of ticket i , P_i , is

$$\begin{aligned} P_i &= (1 - z)(1 - \pi_i) - (1 - z)p_i \left(\frac{(1 - \pi_i)}{\pi_i} \right) \\ &= (1 - z) \left(1 - \pi_i - \frac{p_i - p_i\pi_i}{\pi_i} \right) \\ &= (1 - z) \left(1 - \pi_i + p_i - \frac{p_i}{\pi_i} \right) \end{aligned}$$

The overall expected payout P for the Bookie from Outsiders therefore is

$$\begin{aligned}
P &= P_1 + P_2 \\
&= (1-z) \left(1 - \pi_1 + p_1 - \frac{p_1}{\pi_1} \right) + (1-z) \left(1 - \pi_2 + p_2 - \frac{p_2}{\pi_2} \right) \\
&= (1-z) \left(2 - \pi_1 - \pi_2 + p_1 + p_2 - \frac{p_1}{\pi_1} - \frac{p_2}{\pi_2} \right) \\
&= (1-z) \left(2 - \pi_1 - \pi_2 - p_1 \frac{1-\pi_1}{\pi_1} - p_2 \frac{1-\pi_2}{\pi_2} \right)
\end{aligned}$$

Against the Insider, the Bookie always has $\mathcal{L}z$ of revenue, because the Insider will always bet all of her wealth on the horse that she knows will win. However, the Bookie loses z/π_1 with probability p_1 and z/π_2 with probability p_2 . Therefore the overall expected payout I for the Bookie from the Insider is

$$\begin{aligned}
I &= z - \frac{zp_1}{\pi_1} - \frac{zp_2}{\pi_2} \\
&= z \left(1 - \frac{p_1}{\pi_1} - \frac{p_2}{\pi_2} \right)
\end{aligned}$$

So we can combine these and find that the Bookie's total expected payout is a function of (π_1, π_2) such that

$$\begin{aligned}
f(\pi_1, \pi_2) &= (1-z) \left(2 - \pi_1 - \pi_2 - p_1 \frac{1-\pi_1}{\pi_1} - p_2 \frac{1-\pi_2}{\pi_2} \right) + z \left(1 - \frac{p_1}{\pi_1} - \frac{p_2}{\pi_2} \right) \\
&= 2 + (z-1)(\pi_1 + \pi_2) - z + \frac{(-p_1 z \pi_1 - p_1 + p_1 \pi_1)}{\pi_1} + \frac{(-p_2 z \pi_2 - p_2 + p_2 \pi_2)}{\pi_2} \\
&= 2 + (z-1)(\pi_1 + \pi_2) - z - p_1 z - \frac{p_1}{\pi_1} + p_1 - p_2 z - \frac{p_2}{\pi_2} + p_2 \\
&= 3 + (z-1)(\pi_1 + \pi_2) - 2z - \frac{p_1}{\pi_1} - \frac{p_2}{\pi_2} \quad (p_1 + p_2 = 1)
\end{aligned}$$

and we can calculate the following first and second-order partial derivatives.

$$\begin{aligned}
\frac{\partial}{\partial \pi_1} f(\pi_1, \pi_2) &= z - 1 + \frac{p_1}{\pi_1^2}, & \frac{\partial}{\partial \pi_2} f(\pi_1, \pi_2) &= z - 1 + \frac{p_2}{\pi_2^2} \\
\frac{\partial^2}{\partial \pi_1^2} f(\pi_1, \pi_2) &= -\frac{2p_1}{\pi_1^3} < 0, & \frac{\partial^2}{\partial \pi_2^2} f(\pi_1, \pi_2) &= -\frac{2p_2}{\pi_2^3} < 0
\end{aligned}$$

Notice that the second-order partial derivatives are negative and so f is a concave function of (π_1, π_2) and any stationary points we find will maximise the Bookies expected payout. Setting the first-order equations equal to zero we find that setting

$$\pi_1 = \sqrt{\frac{p_1}{1-z}}, \quad \pi_2 = \sqrt{\frac{p_2}{1-z}}$$

will maximise the expected payout for the Bookie. Note that we only accept bets on Horse i if

$$z < 1 - p_i$$

i.e. the proportion of wealth being bet by Insiders is sufficiently small and $\sqrt{p_i/(1-z)} < 1$. If we suspect the proportion of wealth being bet by Insiders to be too great we get $\sqrt{p_i/(1-z)} \geq 1$ and we set odds greater than 1 which is equivalent to not accepting any bets on this horse (no rational bettor will buy a ticket with a best case return of $\mathcal{L}1$ for more than $\mathcal{L}1$). Shin writes this formally as

$$\pi_i = \min \left\{ 1, \sqrt{p_i/(1-z)} \right\}$$

Let's now graph an example case to see how our ticket prices change with respect to p_1 .

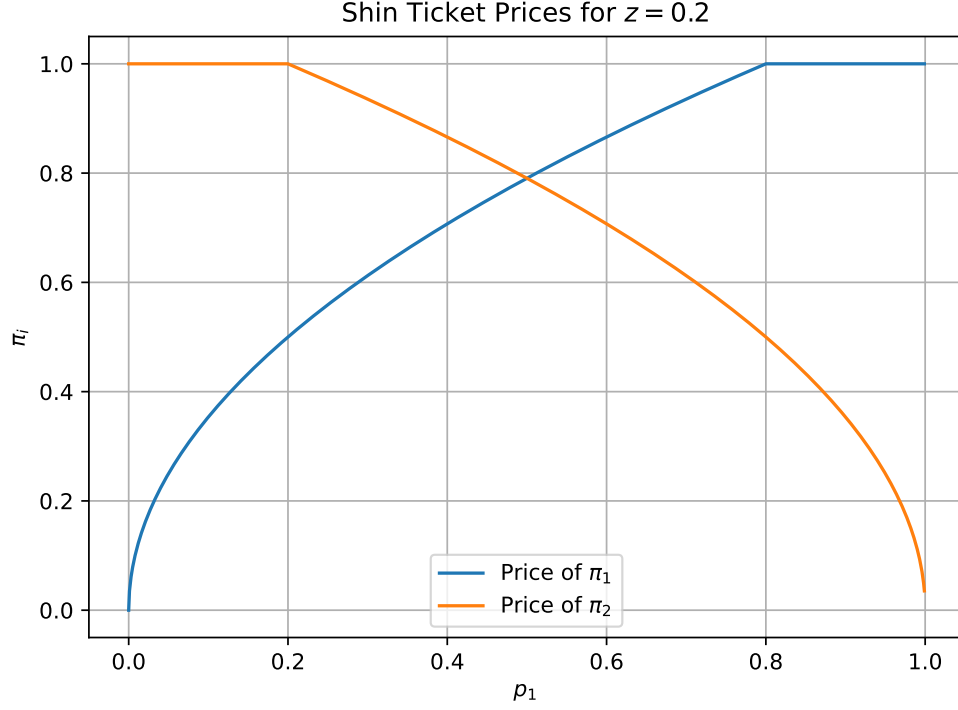


Figure 1: Shin Ticket Prices for $z = 0.2$

We can see that if we believe that 20% of the wealth being gambled comes from the Insider, we stop accepting bets on both horses when $p_1 = 0.8 \implies p_2 = 0.2$. In an even race, we price both tickets at roughly 0.8 which is far higher than the fair price of 0.5. We can see that both lines are concave, which means it is adjusting for the favourite-longshot bias as we intend [12]. This phenomenon sees the favourite priced much more accurately than the longshot, in this case as a result of the Bookie's fear that the Insider will be able to easily exploit longer odds. Let's examine the prices when we believe that there is no Insider trading occurring.

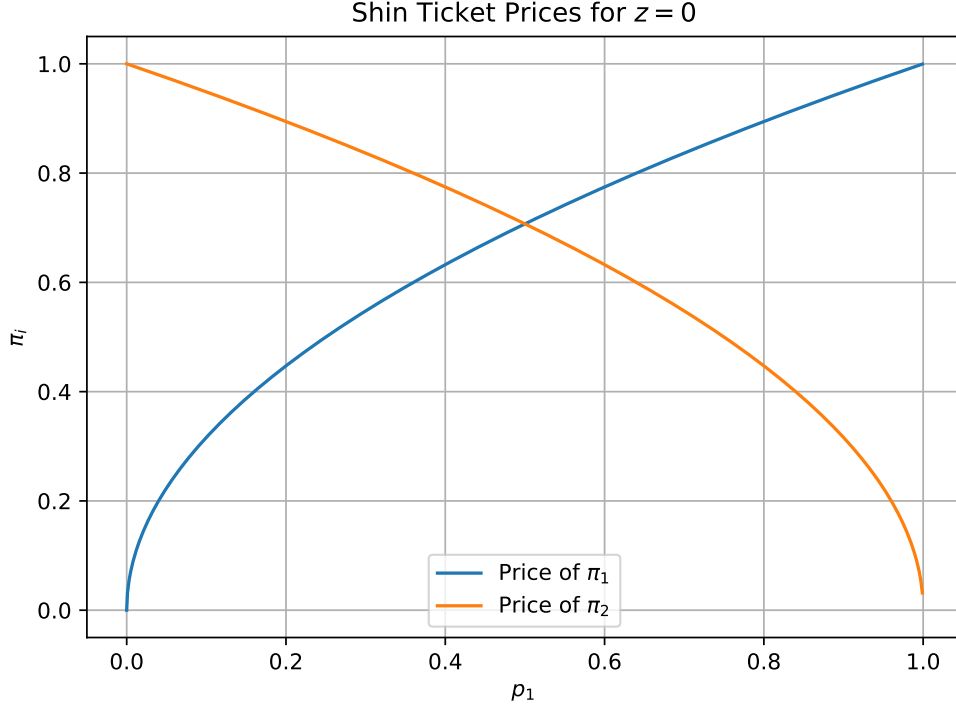


Figure 2: Shin Ticket Prices with no Insider

Here we are happy to accept bets no matter what value of p_1 has been chosen by Nature, and the prices are still concave functions. However, this graph may make us very sceptical of the practicality of using Shin's method in real-life scenarios. This graph suggests that in an even contest, the optimal price of a ticket for either horse is $\sqrt{0.5} \approx \pounds 0.71$! This is equivalent to about 1.41 in decimal odds. For a real life event with $p = 0.5$ such as the coin toss in a Cricket match, bookmakers will often offer 1.9-1.95 odds on either outcome - Shin's price is so far from reality that we may wonder how this method has any use to us.

The reason this result is so far from what we see in reality is that Shin's method assumes that Outsider's beliefs about the value of p_1 are **uniformly distributed** over the interval $[0, 1]$. On a market such as betting on a coin toss, we can see that this assumption is invalid - excluding a small minority who are betting on an irrational belief, most Outsiders will share the belief that $p_1 = 0.5$. If we consider an event like a very evenly matched Darts match between two players A and B , bookmakers often post similar odds to the coin flip example of about 1.9-1.95 for either player to win. In this scenario, there is likely to be more variation in the beliefs of the Outsiders, but to model this variation as uniform over $[0, 1]$ still feels incorrect (we may instead think that q is more likely to come from a $\mathcal{N}(0.5, 0.1^2)$ distribution for example). Secondly, we have modelled the Bookie as monopolistic, which is also wildly inaccurate. In reality, there are many different bookmakers competing with each other to offer the lowest possible prices π_i whilst still maintaining their margin. A rival bookmaker in this example could price the tickets considerably cheaper whilst still having a positive expected profit.

In the next section we will consider a far more realistic case where there are $n \geq 3$ outcomes, and the behaviour of the Bookie and Outsiders is more realistic.

3.2 Shin's Method for $n \geq 3$ Outcomes

In this section we are aiming for a similar outcome to the previous one, with explicit expressions for parameters, and with the formula for π_i exhibiting favourite-longshot bias. We describe the model according to Shin (1993)[13].

In this scenario we model a horse race involving n horses. In a deviation from the previous section, there are now 2 bookmakers, who will compete to be the monopolistic Bookie. One of these bookmakers is the *incumbent* bookmaker, who receives small privileges over the *potential* bookmaker. We have an Insider as previously defined, and a set of n Outsiders. The behaviour of these Outsiders differs from the previous section in that Outsider i believes with certainty that horse i will win. This may seem inaccurate but it still allows us to model divergence in beliefs amongst Outsiders, and for the behaviour of the Outsiders as a collective to reveal true probabilities [13].

As before, there are n tickets with prices $\pi_1, \dots, \pi_n$ that pay out $\mathcal{L}1$ if horse i wins. We still enforce the $0 \leq \pi_i \leq 1$ constraint on ticket prices.

The three stages of this game are now as follows:

- Stage 1. **Bidding:** The two bookmakers bid for monopoly rights to the betting market. These bids are positive real numbers, and they represent the maximum booksum β that they will price their tickets at. The bookmaker who submits the lower number wins the rights to the entire market, meaning we will see fairer odds than we got from Shin (1991). If both bookmakers submit the same bid then the incumbent bookmaker is awarded the rights.
- Stage 2. **Price-Setting:** The winning Bookie now sets their prices π_i subject to the constraints $0 \leq \pi_i \leq 1$ and $\sum_i \pi_i \leq \beta$.
- Stage 3. **Betting:** Nature performs two experiments. The first is the same as in the previous section - Nature picks a winning horse from the Categorical distribution $(p_1, \dots, p_n)$, $p_i > 0$. Nature then chooses exactly one gambler who will place a bet with the Bookie. The Insider is chosen with probability z , and the i -th Outsider is chosen with probability $(1 - z)p_i$. The Outsiders will always bet $\mathcal{L}1$ on the horse they are certain will win, and the Insider, who is permitted to observe the outcome of Nature's first experiment, will always bet $\mathcal{L}1$ on the horse they know will win. The race is then run according to Nature and the Bookie settles the bet with the chosen gambler.

Note that (i) the competition between the bookmakers will drive their expected profit towards 0, and (ii) this model can be generalised to a large number of gamblers without changing the mathematical problem facing the bookmakers [13].

Let's now derive some expressions for the expected revenues and profit for the winning Bookie on each ticket i . Considering only Outsiders betting on horse i , the bookie has

- Expected Revenue of

$$p_i(1 - z)$$

There is a $(1 - z)$ chance that an Outsider is picked as the gambler, and they will bet $\mathcal{L}1$ on ticket i with probability p_i .

- Expected Payout of

$$(1 - z) \left(\frac{p_i^2}{\pi_i} \right)$$

The bookie will have to payout $1/\pi_i$ lots of the stake $(1 - z)p_i$ with probability p_i .

Now considering the Insider, the bookie has

- Expected Revenue of

$$zp_i$$

There is a z chance that the Insider is picked as the gambler, and they will bet $\mathcal{L}1$ on ticket i with probability p_i .

- Expected Payout of

$$z \left(\frac{p_i}{\pi_i} \right)$$

The bookie will have to payout $1/\pi_i$ lots of the stake zp_i with probability 1.

Combining these expressions, and summing over all tickets $i = 1, \dots, n$, we get the following expression for the total Expected Profit (EP) of the Bookie

$$\begin{aligned} EP(\boldsymbol{\pi}) &= \sum_{i=1}^n \{(1-z)p_i + zp_i\} - \sum_{i=1}^n \left\{ (1-z) \frac{p_i^2}{\pi_i} + \frac{zp_i}{\pi_i} \right\} \\ &= \sum_{i=1}^n p_i - \sum_{i=1}^n \frac{zp_i + (1-z)p_i^2}{\pi_i} \\ &= 1 - \sum_{i=1}^n \frac{zp_i + (1-z)p_i^2}{\pi_i} \end{aligned}$$

This results in the following optimisation problem facing the bookmaker when setting their prices $\boldsymbol{\pi}$:

$$\boldsymbol{\pi} = \arg \max_{\boldsymbol{\pi} \in \mathbb{R}^{(+)^n}} \left\{ 1 - \sum_{i=1}^n \frac{zp_i + (1-z)p_i^2}{\pi_i} \right\}$$

subject to the constraints $\sum_i \pi_i \leq \beta$ and $0 \leq \pi_i \leq 1 \quad \forall i$.

We can approach this problem using the method of Lagrange multipliers to find the maxima of $EP(\boldsymbol{\pi})$ subject to the constraints. We define the following Lagrangian

$$\mathcal{L}(\boldsymbol{\pi}, \lambda) = 1 - \sum_{i=1}^n \frac{zp_i + (1-z)p_i^2}{\pi_i} + \lambda \left(\beta - \sum_{i=1}^n \pi_i \right)$$

To find our maxima we require that both

$$\frac{\partial \mathcal{L}}{\partial \pi_i} \mathcal{L}(\boldsymbol{\pi}, \lambda) = 0 \implies \frac{zp_i + (1-z)p_i^2}{\pi_i^2} - \lambda = 0 \quad \text{for } i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} \mathcal{L}(\boldsymbol{\pi}, \lambda) = 0 \implies \beta - \sum_{i=1}^n \pi_i = 0$$

hold. We therefore get the following expression for π_i :

$$\begin{aligned} \lambda &= \frac{zp_i + (1-z)p_i^2}{\pi_i^2} \\ \pi_i &= \sqrt{\frac{zp_i + (1-z)p_i^2}{\lambda}} \end{aligned}$$

So the $\beta - \sum_i \pi_i = 0$ expression becomes:

$$\begin{aligned}
\beta &= \sum_{i=1}^n \pi_i \\
&= \sum_{i=1}^n \sqrt{\frac{zp_i + (1-z)p_i^2}{\lambda}} \\
&= \sqrt{\frac{1}{\lambda}} \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \\
\beta\sqrt{\lambda} &= \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \\
\lambda &= \left(\frac{1}{\beta} \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \right)^2
\end{aligned}$$

And substituting this back into the equation for π_i we get

$$\pi_i = \frac{\beta \sqrt{zp_i + (1-z)p_i^2}}{\sum_{j=1}^n \sqrt{zp_j + (1-z)p_j^2}}$$

Let's substitute this new expression for π_i into $EP(\pi)$:

$$\begin{aligned}
EP(\pi) &= 1 - \sum_{i=1}^n \frac{zp_i + (1-z)p_i^2}{\pi_i} \\
&= 1 - \sum_{i=1}^n \frac{zp_i + (1-z)p_i^2}{\left(\beta \sqrt{zp_i + (1-z)p_i^2} \right) / \left(\sum_{j=1}^n \sqrt{zp_j + (1-z)p_j^2} \right)} \\
&= 1 - \frac{1}{\beta} \sum_{i=1}^n \frac{(zp_i + (1-z)p_i^2) \sum_{j=1}^n \sqrt{zp_j + (1-z)p_j^2}}{\sqrt{zp_i + (1-z)p_i^2}} \\
&= 1 - \frac{1}{\beta} \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \sum_{j=1}^n \sqrt{zp_j + (1-z)p_j^2} \\
&= 1 - \frac{1}{\beta} \left\{ \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \right\}^2
\end{aligned}$$

Given that we have competition between bookmakers, and that the expected payout for the losing bookmaker is 0, $EP(\pi) = 0$ and we can solve for the winning bid β :

$$\beta = \left\{ \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \right\}^2$$

Finally we substitute this expression into the expression for π_i to find

$$\pi_i = \sqrt{zp_i + (1-z)p_i^2} \left(\sum_{j=1}^n \sqrt{zp_j + (1-z)p_j^2} \right)$$

Graphing these prices for the two-horse race example we saw earlier we get the following prices:

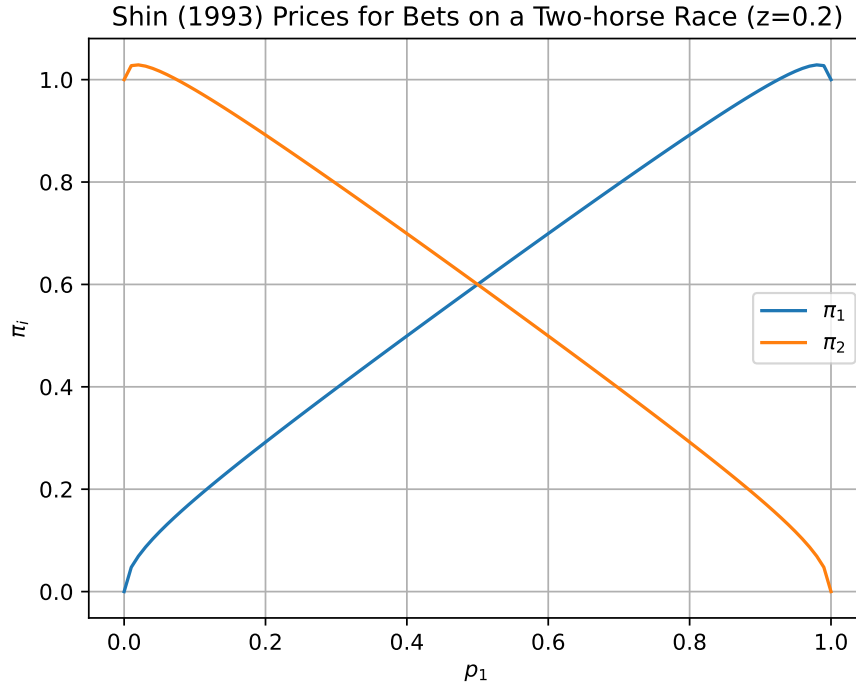


Figure 3: Shin (1993) Ticket Prices with $n = 2, z = 0.2$

We can see that the Bookie stops accepting bets at roughly $p_1 = 0.9$, and if the race is truly fair we offer $\pi_1 = \pi_2 = 0.6$ which is much closer to the sort of overround we see in real-life. The prices also exhibit favourite-longshot bias as we desired. When $z = 0$, $\pi_i = p_i$ and the competition between bookmakers drives prices down to the fair price.

Note that this is an equation for setting prices π given a bookmaker's estimate of z . We now proceed to the problem of inferring $\mathbf{p}$ from observed π .

3.3 Using Shin's Method for Probability Inference

Now we have some understanding of the intuition behind Shin's method, we are interested in how we can use the method to reverse engineer the values of p_i in the equations above. We follow a similar method to that described in Jullien and Salanie (1994) [9]. Recall the equation for π_i

$$\pi_i = \frac{\beta \sqrt{z p_i + (1-z) p_i^2}}{\sum_{j=1}^n \sqrt{z p_j + (1-z) p_j^2}}$$

Let $\Pi = \sum_i \pi_i$. Then $\Pi = \beta$ by definition, and we label the constant $c = \sum_{j=1}^n \sqrt{z p_j + (1-z) p_j^2}$. Using our

expression from the previous section for β we get

$$\begin{aligned}\Pi &= \beta = \left\{ \sum_{i=1}^n \sqrt{zp_i + (1-z)p_i^2} \right\}^2 \\ \Pi &= c^2 \\ \sqrt{\Pi} &= c\end{aligned}$$

So substituting into the initial equation for π_i we have

$$\begin{aligned}\pi_i &= \frac{\Pi \sqrt{zp_i + (1-z)p_i^2}}{\sqrt{\Pi}} \\ &= \sqrt{\Pi} \sqrt{zp_i + (1-z)p_i^2} \\ \frac{\pi_i}{\sqrt{\Pi}} &= \sqrt{zp_i + (1-z)p_i^2}\end{aligned}$$

We now rearrange to get an expression for p_i :

$$\begin{aligned}\frac{\pi_i^2}{\Pi} &= zp_i + (1-z)p_i^2 \\ (1-z)p_i^2 + zp_i - \frac{\pi_i^2}{\Pi} &= 0 \\ p_i &= \frac{-z \pm \sqrt{z^2 + 4(1-z)\left(\frac{\pi_i^2}{\Pi}\right)}}{2(1-z)} \quad (\text{quadratic formula}) \\ p_i &= \frac{-z + \sqrt{z^2 + 4(1-z)\left(\frac{\pi_i^2}{\Pi}\right)}}{2(1-z)} \quad (\text{positive root, } p_i \in [0, 1])\end{aligned}$$

Finally, we need to find an estimate of z for a betting market $\boldsymbol{\pi}$. We can do this by iteratively finding the root z^* of the equation

$$g(z) = \sum_{i=1}^n p_i - 1$$

such that

$$g(z^*) = \sum_{i=1}^n \left\{ \frac{-z^* + \sqrt{z^{*2} + 4(1-z^*)\left(\frac{\pi_i^2}{\Pi}\right)}}{2(1-z^*)} \right\} - 1 = 0$$

We know that $z \in (0, 1)$ by definition, and we use a computational root-finding method to find z^* , which we can then plug back in to our p_i expression to get valid probability values that will sum to 1.

3.4 Proof that Shin's and Additive Methods are Equivalent for n=2 Outcomes

When outcomes are binary, the method above is actually equivalent to the Additive method we saw in Section 2. Clarke et al. (2017) [3] presents two possible proofs for this result. Here we present a cleaner

proof beginning from an expression we found in the previous section

$$\frac{\pi_i^2}{\Pi} = zp_i + (1-z)p_i^2$$

Let's write this for $i = 1, 2$ and subtract the two expressions:

$$\begin{aligned}\frac{\pi_1^2}{\Pi} - \frac{\pi_2^2}{\Pi} &= zp_1 + (1-z)p_1^2 - zp_2 - (1-z)p_2^2 \\ \frac{\pi_1^2 - \pi_2^2}{\Pi} &= z(p_1 - p_2) + (1-z)(p_1^2 - p_2^2) \\ \frac{(\pi_1 + \pi_2)(\pi_1 - \pi_2)}{\Pi} &= z(p_1 - p_2) + (1-z)(p_1 - p_2)(p_1 + p_2) \\ \frac{\Pi(\pi_1 - \pi_2)}{\Pi} &= z(p_1 - p_2) + (1-z)(p_1 - p_2) \quad [p_1 + p_2 = 1] \\ \pi_1 - \pi_2 &= (p_1 - p_2)(z + 1 - z) \\ &= p_1 - p_2\end{aligned}$$

So if we now solve

$$\begin{cases} p_1 + p_2 = 1 \\ p_1 - p_2 = \pi_1 - \pi_2 \end{cases}$$

We find that

$$p_1 = \frac{\pi_1 - \pi_2 + 1}{2} = \frac{\pi_1 - \Pi + \pi_1 + 1}{2} = \pi_1 - \frac{\Pi - 1}{2}$$

and the equivalent is true for p_2 . This is exactly the same as the additive method where $\pi_i = 1/o_i$ and $\Pi = \beta$ is the overround.

4 Regression Methods

Our regression methods will differ from the methods we have previously seen in that we will be building models that are trained on historical data. This means that in order for these method to be useful, we need access to a large dataset of historical match results and bookmaker odds. For the methods described below we will lay out the mathematical intuition behind these models and then fit them to example data.

4.1 Linear Regression

We will first define a simple linear regression model of the form

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

where $\mathbf{X}$ is the $(N \times 2)$ design matrix, with $\mathbf{X}_0 = [1, \dots, 1]^T$ being the intercept term, and $\mathbf{X}_1$ being the column of bookmaker implied probabilities. $\mathbf{y}$ is the $(N \times 1)$ response vector, which represents the probabilities we are trying to find. β is the (2×1) vector of regression coefficients, and ϵ is Gaussian noise. We perform Ordinary Least Squares (OLS) regression to find $\hat{\beta}_{OLS}$.

Note that the predictors of this model are not exact implied probabilities, but bins of similar probabilities, with the corresponding response being the proportion of the time these probabilities corresponded to a winning bet. The size of these bins is a hyper-parameter that we can choose. For example we may choose to split the implied probabilities into 20 bins of size 0.05 that span the interval $[0, 1]$ We then take the average implied probability within that bin to be the predictor.

Our model is attempting to find the value

$$\hat{\beta}_{OLS} = \arg \min_{\beta \in \mathbb{R}^p} RSS(\beta)$$

where

$$RSS(\beta) := \|\mathbf{y} - \mathbf{X}\beta\|_2^2 = \sum_{i=1}^N (y_i - x_i^T \beta)^2$$

We assume that $\mathbf{X}$ has full column rank in this solution so that we can invert $\mathbf{X}$. Let's expand $RSS(\beta)$:

$$\begin{aligned} RSS(\beta) &= \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \\ &= (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}^T \mathbf{y} - (\mathbf{X}\beta)^T \mathbf{y} - \mathbf{y}^T (\mathbf{X}\beta) + (\mathbf{X}\beta)^T (\mathbf{X}\beta) \\ &= \mathbf{y}^T \mathbf{y} - 2\beta^T \mathbf{X}^T \mathbf{y} + \beta^T \mathbf{X}^T \mathbf{X} \beta \end{aligned}$$

Differentiating we get

$$\frac{\partial}{\partial \beta} RSS(\beta) = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \beta$$

and setting to zero at $\hat{\beta}_{OLS}$:

$$\begin{aligned} \left. \frac{\partial}{\partial \beta} RSS(\beta) \right|_{\beta=\hat{\beta}_{OLS}} &= -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \hat{\beta}_{OLS} = 0 \\ 2\mathbf{X}^T \mathbf{X} \hat{\beta}_{OLS} &= 2\mathbf{X}^T \mathbf{y} \\ \hat{\beta}_{OLS} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \end{aligned}$$

Finally we check that this point is in fact a minimum:

$$\left. \frac{\partial^2}{\partial \beta^2} RSS(\beta) \right|_{\beta=\hat{\beta}_{OLS}} = 2\mathbf{X}^T \mathbf{X}$$

which is positive definite for all β and so we have found the analytical solution to this minimisation problem

$$\hat{\beta}_{OLS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

but we will be using software to compute the OLS estimator for us.

Consider an example using English Premier League Data from 2006-2026. We have a dataset with 7210 match results, and the Bet365 odds offered on each of the 3 outcomes {Home, Draw, Away}. Our data for Home Win probabilities is structured as follows

Implied Prob. Bin	Avg. Implied Prob.	Empirical Prob.	Count
(0, 0.05]	0.04555	0.00000	4
(0.05, 0.1]	0.08576	0.06897	116
$\vdots$	$\vdots$	$\vdots$	$\vdots$
(0.85, 0.9]	0.86997	0.92361	144
(0.9, 0.95]	0.91525	0.92000	50

Table 1: Implied vs Empirical Probabilities for Home Win odds: PL 2006-2026

The data is structured similarly for both the Draw and Away Win outcomes. Let's examine plots of the three models below.

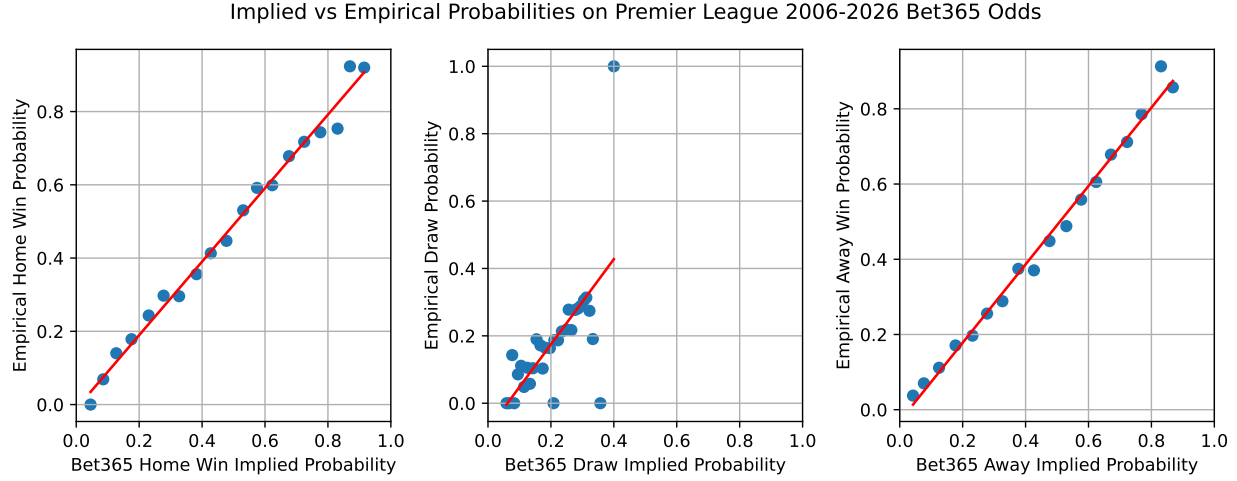


Figure 4: Bet365 Implied vs Empirical Probabilities

Note that the bins for Draw probabilities were readjusted to reflect the fact that the Draw was never priced at more than 2.5 in our dataset. We instead took 50 bins from the interval $[0, 0.4]$.

We can see that for Home and Away Wins, these models are already fairly accurate, with some inaccuracies at the tails but otherwise a strong fit to the data. Given the volume of money that is staked on the English Premier League and the amount of data collected by analysts it makes sense that this market is very efficient - this is not always the case in markets for other leagues or other sports. The model struggles however to deal with the Draw. In order to use this model for probability inference on unseen data, we use the parameters to form three linear equations to find $\mathbf{p} = (p_H, p_D, p_A)$ from $\mathbf{o} = (o_H, o_D, o_A)$. From our code we get

$$p_H = 1.00339o_H - 0.01111$$

$$p_D = 1.26486o_D - 0.07841$$

$$p_A = 1.04191o_A - 0.03036$$

If we wanted to enforce the sums of these probabilities to be 1, we could set $p_D = 1 - p_H - p_A$.

4.2 Polynomial Regression

Polynomial regression takes a very similar approach but adds in polynomial terms for predictors. We will consider the case where the degree of our polynomial $d = 2$ but for some applications it may be more suitable to use cubic regression. It is unlikely that anything beyond $d = 3$ would ever be necessary for this case. In the polynomial regression model, for every existing predictor we add columns to the design matrix representing different powers of that predictor, and we therefore require extra regression coefficients for each of these columns. In our case, we have

$$p_i = \beta_0 + \beta_1 o_i + \beta_2 o_i^2 + \epsilon$$

We estimate β using the same OLS method as before and get the following model fits to our Premier League data.

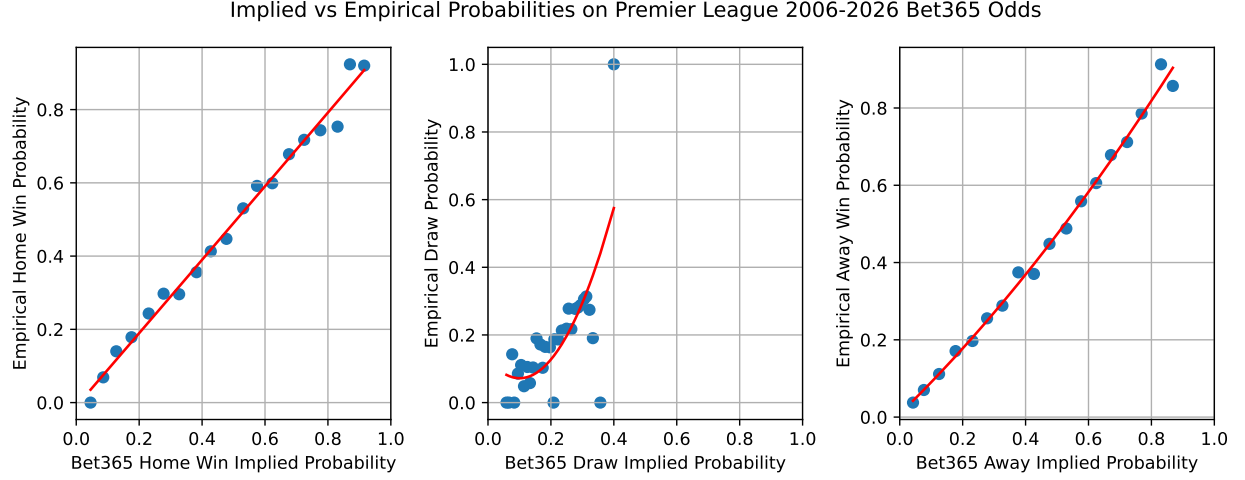


Figure 5: Bet365 Implied vs Empirical Probabilities

And the following equations for $\mathbf{p}$:

$$p_H = 0.99830o_H + 0.00532o_H^2 - 0.01028$$

$$p_D = -1.13381o_D + 5.62037o_D^2 + 0.12922$$

$$p_A = 0.78956o_A + 0.27896o_A^2 + 0.00839$$

4.3 Logistic Regression

In the case of $n = 2$ outcomes, we can use a type of Generalised Linear Model (GLM) called a Logistic Regression to find probability estimates. In this model we construct the predictor vector $\mathbf{q}_i = (1, q_{i1}) = (1, 1/o_{i1})$ which contains an intercept term and the implied probability of the odds o_{i1} offered on outcome 1 in event i . We also define

$$y_i = \begin{cases} 1 & \text{if outcome 1 occurred in event } i \\ 0 & \text{if outcome 2 occurred in event } i \end{cases}$$

as a binary outcome variable. In standard GLM notation we have

$$\mathbb{E}(y_i | \mathbf{q}_i = \mathbf{q}) = \mu(\mathbf{q})$$

$$\mu(\mathbf{q}) = g^{-1}(\mathbf{q}^T \beta)$$

where $\mu : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the regression function and $g : \mathbb{R} \rightarrow \mathbb{R}$ is the link function. For logistic regression we assume that

$$y_i | \mathbf{q}_i = \mathbf{q} \sim \text{Bernoulli}(p(\mathbf{q}))$$

where $p(\mathbf{q})$ is our regression function that we are trying to estimate. We choose our link function such that

$$\begin{aligned} g(x) &= \log\left(\frac{x}{1-x}\right) \\ g(p(\mathbf{q})) &= \log\left(\frac{p(\mathbf{q})}{1-p(\mathbf{q})}\right) \\ \mathbf{q}^T \beta &= \log\left(\frac{p(\mathbf{q})}{1-p(\mathbf{q})}\right) \\ p(\mathbf{q}) &= \frac{1}{1 + e^{-\mathbf{q}^T \beta}} \\ &= \frac{1}{1 + e^{-(\beta_0 + \beta_1 q_1)}} \end{aligned}$$

We now perform MLE to find $\hat{\beta}_{MLE}$ which we plug-in to find $\hat{p}(\mathbf{q})$ which is the function we can use to infer valid probabilities. From the conditional distribution of y_i we have

$$\mathbb{P}(y_i = y | \mathbf{q}_i = \mathbf{q}) = p(\mathbf{q})^y (1 - p(\mathbf{q}))^{1-y}$$

We then get the following likelihood function of β :

$$L(\beta; y_i, \mathbf{q}_i, i = 1, \dots, N) = \prod_{i=1}^N p(\mathbf{q}_i)^{y_i} (1 - p(\mathbf{q}_i))^{1-y_i}$$

and the log-likelihood:

$$\ell(\beta; \cdot) = \sum_{i=1}^N y_i \log(p(\mathbf{q}_i)) + (1 - y_i) \log(1 - p(\mathbf{q}_i))$$

So

$$\hat{\beta}_{MLE} = \arg \max_{\beta \in \mathbb{R}^2} \left\{ \sum_{i=1}^N y_i \log(p(\mathbf{q}_i)) + (1 - y_i) \log(1 - p(\mathbf{q}_i)) \right\}$$

There is no analytical solution to this problem so we use computational methods to compute $\hat{\beta}_{MLE}$ and subsequently $\hat{p}(\mathbf{q})$. If we are then presented with a new set of odds $\mathbf{o} = (o_1, o_2)$, we can calculate $p_1 = \hat{p}(1, o_1)$ and set $p_2 = 1 - p_1$ since the outcomes are binary.

4.4 Ordered Logistic Regression

We can expand this method into $n \geq 3$ **ordered** outcomes using an ordered logistic regression model. In this section n refers to the number of outcomes, and N refers to the number of events in our dataset. If we have an event with n ordered outcomes, let $Y \in \{1, \dots, n\}$ be the outcome of the event, for example these could map to $Y \in \{\text{Home Win, Draw, Away Win}\}$ in a football match, as this is the logical ordering of the outcomes from the home team's perspective (best outcome to worst outcome). Then modelling

$$\mathbb{P}(Y \leq k), \quad k = 1, \dots, n-1$$

allows us to model all of the individual outcome probabilities using

$$\mathbb{P}(Y = k) = \mathbb{P}(Y \leq k) - \mathbb{P}(Y \leq k-1)$$

where we set $\mathbb{P}(Y \leq 0) = 0$, and $\mathbb{P}(Y \leq n) = 1$. We can easily see that this will result in probabilities that sum to 1:

$$\sum_{k=1}^n \mathbb{P}(Y = k) = \mathbb{P}(Y \leq n) - \mathbb{P}(Y \leq 0) = 1 - 0 = 1$$

We model these cumulative probabilities using $n - 1$ logistic functions

$$\mathbb{P}(Y_i \leq k | \mathbf{q}_i) = \frac{1}{1 + \exp(-(\alpha_k - \beta^T \mathbf{q}_i))} \quad \text{for } k = 1, \dots, n - 1$$

where $\beta \in \mathbb{R}^n$, and $\alpha_1 < \dots < \alpha_{n-1}$ are threshold parameters that are analogous to the intercept values for each cumulative boundary k . Note that α is different for each outcome, but β is constant for every outcome - this is the proportional odds model from McCullagh (1980) [10]. We will estimate these parameters in the same way as logistic regression, by maximising the log-likelihood

$$\ell(\alpha, \beta) = \sum_{i=1}^N \log(\mathbb{P}(Y_i = y_i | \mathbf{q}_i))$$

to find $\hat{\beta}_{MLE}, \hat{\alpha}_{MLE}$.

This has no closed form solution so we will use software to find the values that maximise this function.

5 Data

Below we outline the data we will be using to test our methods. The exact format of the data varies between sports, but in general the data will consist of decimal odds for all the different outcomes, and the actual outcome of the event.

When we work with our regression models, it is important that we **split the data** into training and testing sets to prevent overfitting. We will then compare all of the methods using only the testing set, so that all of our methods are being compared on the exact same data. The only regression methods we will present results for here are the logistic and ordered logistic regression methods, as these are the methods where we guarantee that $\sum_i p_i = 1$.

5.1 Data for $n = 2$ Outcomes

One of our datasets for binary outcomes will come from Men's Grand Slam tennis matches in the 2016-2026 period. This dataset consists of 5056 valid matches and odds for either player to win the match. We will use the Bet365 closing odds as the prices. The average overround in this dataset is roughly 5%. Data from <http://tennis-data.co.uk/data.php>

Our other dataset consists of football matches in the Italian Serie A between 2016-2026. This dataset contains 2536 matches with scorelines and odds on the Over/Under 2.5 Goals market. This is a market on the total number of goals that will be scored in the match, and the spread of these odds are generally much tighter than the odds in our tennis dataset (massive outsiders are fairly common in tennis but it's never that unlikely for there to be over or under 2.5 goals in a match). Again we use the Bet365 closing odds as the prices. The average overround in this dataset is roughly 5.4%. Data from <https://www.football-data.co.uk/italym.php>

We will use this data to compare the following methods:

- Multiplicative method
- Power Method
- Shin's Method (equivalent to Additive for $n = 2$)
- Logistic Regression

5.2 Data for $n \geq 3$ Outcomes

We use the same Serie A dataset as described above but we are now betting on the 1X2 match outcome market, with the Bet365 closing odds for Home Win, Draw, Away Win as the prices. There are more matches with data available on this market, so we have 3677 matches in this dataset. The average overround in this dataset is roughly 5.2%.

To examine how our methods perform when there are a larger number of possible outcomes, we also have a dataset of horse racing results from races taking place in 2018 and 2019. There are 34712 races in this dataset, with an average of 11 runners in each race and an average overround of just over 20%. Data from <https://www.kaggle.com/datasets/hwaitt/horse-racing>.

We will use this data to compare the following methods:

- Additive Method
- Multiplicative method
- Power Method
- Shin’s Method
- Ordered Logistic Regression

6 Proper Scoring Rules for Assessing Probabilistic Accuracy

Scoring rules are a popular method of assessing how good a probability forecast is. Assume we have a future event Y with a finite set of $\mathcal{Y}$ possible outcomes. For example, in this report, we may have $\mathcal{Y} = \{\text{Home Win, Draw, Away Win}\}$.

Let $\mathcal{F}$ be the set of all possible probability distributions over $\mathcal{Y}$. Then we define a **scoring rule** to be a function $S : \mathcal{F} \times \mathcal{Y} \rightarrow \mathbb{R}$. Let the **Expected Score** of a scoring rule with true probability distribution p , and forecasted probability distribution q , be defined as

$$S(q, p) = \mathbb{E}_p(S(q, Y)) = \sum_y p(y)S(q, y)$$

A scoring rule is a **proper scoring rule** if this expected score is maximised by the true probability distribution. I.e. if the true probability distribution is p then for all probability distributions $q \in \mathcal{F}, q \neq p$,

$$S(p, p) \geq S(q, p)$$

It is a strictly proper scoring rule if this inequality is strict.

Note that there are a wide range of proper scoring rules available, and many of these are negatively-orientated so that scores closer to 0 are better. The definitions above still hold true for these scoring rules if we flip the sign of the inequality.

In the *Results* section of this report we will use three proper scoring rules to compare the performance of our different methods. The rest of this section will outline the definitions and theory behind these scoring rules.

6.1 Brier Score

The Brier score is a strictly proper scoring rule that is analogous to the mean squared error of a prediction. As defined in Brier (1950) [1], we calculate the Brier score using

$$BS = \frac{1}{N} \sum_{t=1}^N \sum_{i=1}^R (q_{ti} - o_{ti})^2$$

where R is the number of possible outcomes available, N the number of events, q_{ti} the forecasted probability of outcome i in event t , and o_{ti} a binary outcome representing whether outcome i occurred in event t (0 if it didn't happen, 1 if it did). In the case where $R = 2$ (binary outcomes), we use a common simplification of this score

$$BS = \frac{1}{N} \sum_{t=1}^N (1 - q_t)^2$$

where q_t is the predicted probability of the outcome that **did** actually happen in event t . Intuitively this makes sense as we want to minimise how far each predicted probability was from the ideal prediction of 1 for the outcome that occurred.

We will now proceed to show that the Brier Score is a strictly proper scoring rule. Consider the case where we have $N = 1$ events, outcomes occur with true probability $p = (p_1, \dots, p_R)$, and we have forecasted a probability distribution of outcomes $q = (q_1, \dots, q_R)$, $q \neq p$. Let $o(Y)$ be a one-hot vector where $o_i(Y)$ is 1 if outcome i occurred and 0 otherwise. We can write the Brier Score as

$$S(f, Y) = \sum_{i=1}^R (q_i - o_i(Y))^2$$

Given that only one value of $o_i(Y)$ is non-zero, we can expand this to

$$S(f, Y) = \sum_{i=1}^R q_i^2 - 2q_Y + 1$$

Taking expectations over the true underlying distribution p we get

$$\begin{aligned} \mathbb{E}_p(S(f, Y)) &= \mathbb{E}_p \left(\sum_{i=1}^R q_i^2 \right) - \mathbb{E}_p(2q_Y) + \mathbb{E}_p(1) \\ &= \sum_{i=1}^R q_i^2 - 2 \sum_{i=1}^R p_i q_i + 1 \\ &= \left(\sum_{i=1}^R \{q_i^2 - p_i q_i + p_i^2\} \right) + \left(1 - \sum_{i=1}^R p_i^2 \right) \quad (+ \sum_i p_i^2 - \sum_i p_i^2 = 0) \\ &= \sum_{i=1}^R (q_i - p_i)^2 + \left(1 - \sum_{i=1}^R p_i^2 \right) \end{aligned}$$

Since $1 - \sum_i p_i^2$ is constant, we want to minimise the following function

$$f(q) = \sum_{i=1}^R (q_i - p_i)^2$$

with respect to q . We also have a constraint $\sum_i q_i = 1$. We set up the following Lagrangian

$$L(q, \lambda) = \sum_{i=1}^R (q_i - p_i)^2 + \lambda \left(\sum_{i=1}^R q_i - 1 \right)$$

Differentiating with respect to q_i gives us

$$\frac{\partial L}{\partial q_i} = 2(q_i - p_i) + \lambda$$

and setting the derivative equal to 0 at the stationary point gives us

$$q_i = p_i - \frac{\lambda}{2}$$

Summing over all of the q_i 's we get

$$\begin{aligned} \sum_{i=1}^R q_i &= \sum_{i=1}^R p_i - \sum_{i=1}^R \frac{\lambda}{2} = 1 \\ \implies 1 - \frac{R\lambda}{2} &= 1 \implies \lambda = 0 \end{aligned}$$

So we get that there is a stationary point at $\lambda = 0 \implies q_i = p_i$.

Let us finally check that this is a minimum. For $i \neq j$,

$$\begin{aligned} \frac{\partial L}{\partial q_i \partial q_j} &= 0 \\ \frac{\partial L}{\partial q_i^2} &= 2 \end{aligned}$$

So the Hessian matrix of this function is a diagonal matrix with all diagonal entries equal to 2. This means our Hessian matrix is positive definite and we can conclude that there is a unique minimiser of the Brier Score at $q = p$, and so the Brier Score is a strictly proper scoring rule for $N = 1$. This result easily generalises to $N > 1$ as the number of events N has no effect on the minimisation problem we are trying to perform - we are just performing the minimisation problem above N times for different true and forecasted distributions p and q , and we will conclude that $q_{ti} = p_{ti}$ is the unique minimiser when considering events $t = 1, \dots, N$ and outcomes $i = 1, \dots, R$.

6.2 Log-Loss Score

If we observe the outcomes of $t = 1, \dots, N$ events, each with forecasted probability distributions $q_t = (q_{t1}, \dots, q_{tR})$, then we can define the log-loss scoring rule as

$$LogLoss = -\frac{1}{N} \sum_{t=1}^N \log q_{ti}$$

where outcome i is the outcome we observed in event t . Note we are using the natural logarithm here. We can see that this scoring rule will punish models that assigned small probabilities to events that occurred much more harshly than the Brier score. If an outcome occurs when we predicted it had a 0.04 probability of occurring, then we get

$$BS = (0.04 - 1)^2 \approx 0.92$$

$$\text{LogLoss} = -\log(0.04) \approx 1.40$$

and this difference gets larger as q_i gets smaller. This helps us to assess how well models are predicting outcomes in the tails of the probability distribution.

To prove this is a proper scoring rule define the expected score as

$$S(q, p) = \mathbb{E}_p(S(q, Y)) = \sum_y p(y) \log q(y)$$

To prove this is a strictly proper scoring rule, we need to show that for any $q \neq p$,

$$S(p, p) - S(q, p) > 0$$

We begin with

$$\begin{aligned} S(p, p) - S(q, p) &= \sum_y p(y) \log p(y) - \sum_y p(y) \log q(y) \\ &= \sum_y p(y) (\log p(y) - \log q(y)) \\ &= \sum_y p(y) \log \frac{p(y)}{q(y)} \end{aligned}$$

This is the KL-Divergence or relative entropy of p from q , and can be interpreted as “the expected excess surprisal from using approximation q instead of p when the actual is p ”[4]. We know that this value is non-negative, and equal to zero if and only if $p = q$ [4], thus completing the proof.

6.3 Ranked Probability Score

First presented by Epstein (1969) [5], the Ranked Probability Score is a scoring rule that allows us to consider the distance between outcomes, and adjust our score according to that distance. For example, in a football context, if the away team wins the match, we may want to penalise models that predicted a home win more strongly than a model that predicted a draw, as a home win is “further” from an away win than a draw. Similarly to the Ordered Logistic Regression section earlier, we use the idea of cumulative probabilities here.

For a single event, we have

$$RPS = \frac{1}{r-1} \sum_{i=1}^r \left(\sum_{j=1}^i q_j - \sum_{j=1}^i o_j \right)^2$$

where r is the number of outcomes, $\mathbf{q}$ are our probability forecasts, and o_j is the binary outcome variable we have seen before. We order $\mathbf{q}$ and $\mathbf{o}$ in a way that reflects the order of the outcomes. For a set of N events we define the RPS of the predictions as the average RPS of the individual event predictions:

$$RPS = \frac{1}{N} \sum_{t=1}^N \left\{ \frac{1}{r-1} \sum_{i=1}^r \left(\sum_{j=1}^i p_{tj} - \sum_{j=1}^i o_{tj} \right)^2 \right\}$$

Aside from football match outcome betting, RPS can be applied to any market where outcomes have a natural ordering such as betting on exact scorelines in tennis matches (a 2-1 win is closer to a 2-1 loss than a 2-0 loss), or finishing position in a golf tournament (A top 10 finish is closer to a top 5 finish than a top 50 finish). Note that in the case $r = 2$, RPS reduces to the Brier Score. RPS has been proved to be a proper scoring rule [7].

7 Results

7.1 Results on $n = 2$ Outcomes

We begin by presenting results for the 4 relevant methods on our Men’s Grand Slam Tennis dataset. This data was split into an 80:20 train/test split and so we are testing results on 1012 matches of this dataset.

Method	Brier Score	Log-Loss
Multiplicative	0.1641	0.4987
Power	0.1635	0.4968
Shin	0.1636	0.4970
Logistic Regression	0.1667	0.5099

Table 2: Results on Men’s Grand Slam Tennis Data

We can see that the Power and Shin methods perform the best, with logistic regression performing the worst by a significant margin.

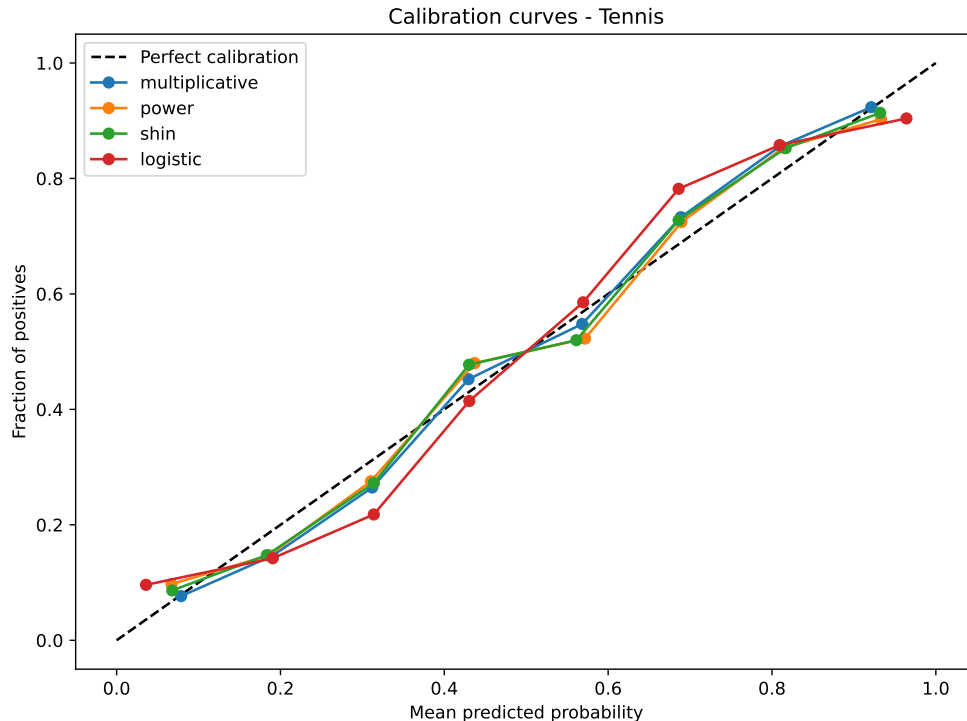


Figure 6: Calibration Curves on Tennis Grand Slam Data

Our calibration curves are generally good fits, with the logistic regression curve straying furthest from the diagonal and the other three methods staying fairly smooth. The flattening of the curves at both tails shows us the overconfidence of our models, particularly logistic regression.

We now turn our attention to the Over/Under 2.5 Goals Market data in the Italian Serie A, where the odds are much more concentrated around a 50% implied probability. This dataset has 2536 valid matches from 2016-2026 in the Italian Serie A. We use a 75:25 train/test split and so we get results on 634 matches.

Method	Brier Score	Log-Loss
Multiplicative	0.2382	0.6693
Power	0.2378	0.6685
Shin	0.2380	0.6688
Logistic Regression	0.2394	0.6717

Table 3: Results on Over/Under 2.5 Goals Dataset

Again the Power method performs best, with the Shin method close behind. Logistic regression again performs worst although still provides a fair amount of predictive power - a naive model that gives predictions of 50% for all outcomes will receive

$$BS = (0.5 - 1)^2 = 0.25$$

$$LogLoss = -\log(0.5) \approx 0.693$$

so these are useful benchmarks to make sure our models working properly.

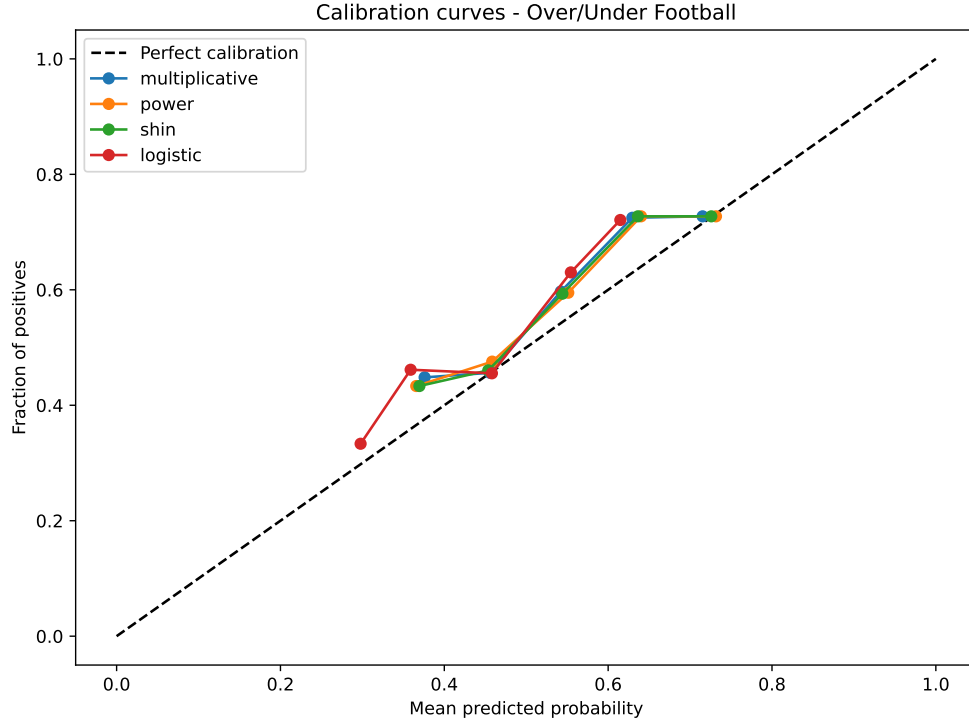


Figure 7: Calibration Curves on Over/Under 2.5 Goals Dataset

We notice that the calibration curve for logistic regression is less smooth than the other three methods again, and the Shin and power curves stay closest to the diagonal.

7.2 Results on $n \geq 3$ Outcomes

We now apply our methods to data on the 1X2 market in the Serie A from 2016-2026. There are 3677 matches in this dataset, and with a 75:25 train/test split we have a testing set of 919 matches. We present the accuracy scores below.

Method	Brier Score	Log-Loss	RPS
Additive	0.5643	0.9532	0.3749
Multiplicative	0.5647	0.9539	0.3754
Power	0.5643	0.9530	0.3749
Shin	0.5644	0.9533	0.3750
Ordered Logistic	0.5621	0.9482	0.3740

Table 4: Results on Serie A 1X2 Market

Unlike the results we have seen thus far, here the regression model trained on the data performs the best, by a significant margin. Of the non-regression methods, the Power method performs best, slightly better than the Additive and Shin methods, and considerably better than the Multiplicative method.

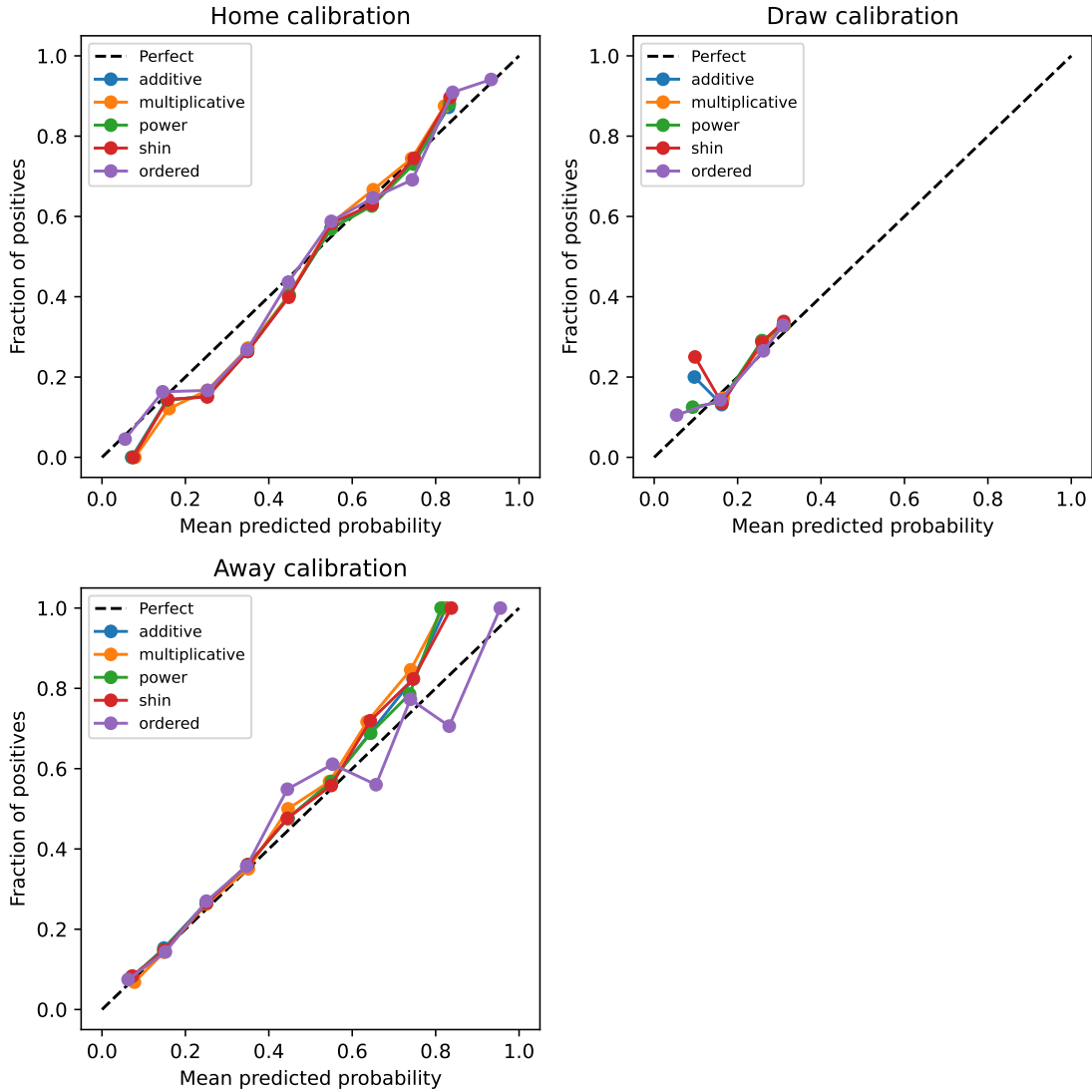


Figure 8: Calibration Curves on Serie A 2016-2026 1X2 Market

We see very strong fits from the methods in general, with the non-regression methods tending to overrate the

chances of outsiders when playing at home more than the ordered regression model. The largest calibration errors are likely occurring due to a lack of samples - away teams aren't often given more than an 80% chance of winning and the Draw rarely has less than a 10% probability.

We pause here to note that although the regression result here is impressive, this model has been trained to learn the pattern between odds and probabilities in the Italian Serie A specifically. We don't expect these results to be portable between leagues. Below we present results where we test this regression model on a dataset of 1510 matches played in the English Championship between 2023-2026, with an average overround of 6.33%.

Method	Brier Score	Log-Loss	RPS
Additive	0.6181	1.0284	0.4267
Multiplicative	0.6180	1.0286	0.4267
Power	0.6182	1.0284	0.4266
Shin	0.6181	1.0284	0.4267
Ordered Logistic (trained on Serie A data)	0.6197	1.0312	0.4276

Table 5: Results on English Championship 1X2 Market

The regression model still performs reasonably well, but is performing worse than the models that haven't been trained on any data.

Finally we present a set of results for markets with lots of possible outcomes. This horse racing dataset has a variable number of horses per race and no natural ordering of the outcomes, so we will not be fitting a regression model to this data, nor will we be able to use the Ranked Probability Score. To be specific we have the finishing positions of the horses in 34712 horse races and the pre-race odds for each horse to win the race. There were an average of 11 horses running in each race. This dataset will give us a feel for how well these methods can handle a wide range of implied probability distributions.

Method	Brier Score	Log-Loss
Additive	0.7801	1.8838
Multiplicative	0.7806	1.8206
Power	0.7794	1.8154
Shin	0.7794	1.8143

Table 6: Results on 2018-2019 Horse Racing Data

On this data, Shin's method performs the best, slightly outperforming the Power method. Notice that although the Additive method outperforms the Multiplicative in Brier Score, its Log-loss is very high due to its potential to assign incredibly small probabilities discussed in section 2. Given that Shin's method originates in a horse-racing scenario, it is reassuring to see that it performs best on this dataset.

8 Conclusion

Betting markets represent one of the most valuable sources of probabilistic information available to sports analysts, modellers, and bettors alike. The combination of very high market liquidity and powerful in-house sports models mean that market odds encode a highly accurate forecast of any sporting event. However, the overround built into every bookmaker's prices creates a barrier between us and this information - the probabilities implied by the odds are not valid probabilities. This report has discussed a range of methods

for removing this overround, from simple rescaling methods through to Shin’s hypothetical model of insider trading, and finally to regression models that learn the relationship between odds and outcomes from historical data.

Using proper scoring rules to assess the performance of these methods, the main finding from this report is that the best method for removing bookmaker overround depends on the market you are looking at. For binary outcome markets with low margin, we found that all methods perform similarly, with the Power method just outperforming our other methods. We also showed that for binary markets, Shin’s method reduces exactly to the Additive method, providing a theoretical explanation for why all methods converge in performance for $n = 2$. A surprising result in this report is that the Multiplicative method, one of the most popular methods due to its simplicity, consistently performed worse than all of our other methods.

On match outcome markets, the Power, Additive, and Shin methods all performed reasonably well, but were outperformed by an Ordered Logistic Regression model fit on historical data. Finally, in markets with a large number of outcomes available, and a higher margin built-in to the odds, the Shin method was the best performing method.

It is useful to know that in scenarios where we have access to lots of historical data, we can use regression methods to infer probabilities from the odds we are being offered. However, we may not always have access to this data, and these models are relevant only to the markets they were trained on - a model trained on the relationship between odds and probabilities in the Serie A does not perform well when applied to the English Championship. This suggests that regression models capture market-specific pricing patterns rather than a universal relationship between odds and probabilities.

The Power and Shin methods performed consistently well, but the true value of these methods is that they don’t require any historical data to be effective. They are able to apply the favourite-longshot bias implicitly and produce accurate estimates with no prior knowledge required. Shin’s thought experiment produces an elegant solution to this problem that performs consistently well for any value of n without any need to train a model or even observe historical data. In scenarios where data is scarce or not publicly available the Shin and Power methods are a valuable tool that allow us to access the true underlying probability forecasts.

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